Single Port Ram Project

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Overview

- instance single port ram & read data
- this is ETRI W6 DAY 2 ASSIGNMENT1.

Features

Constraints

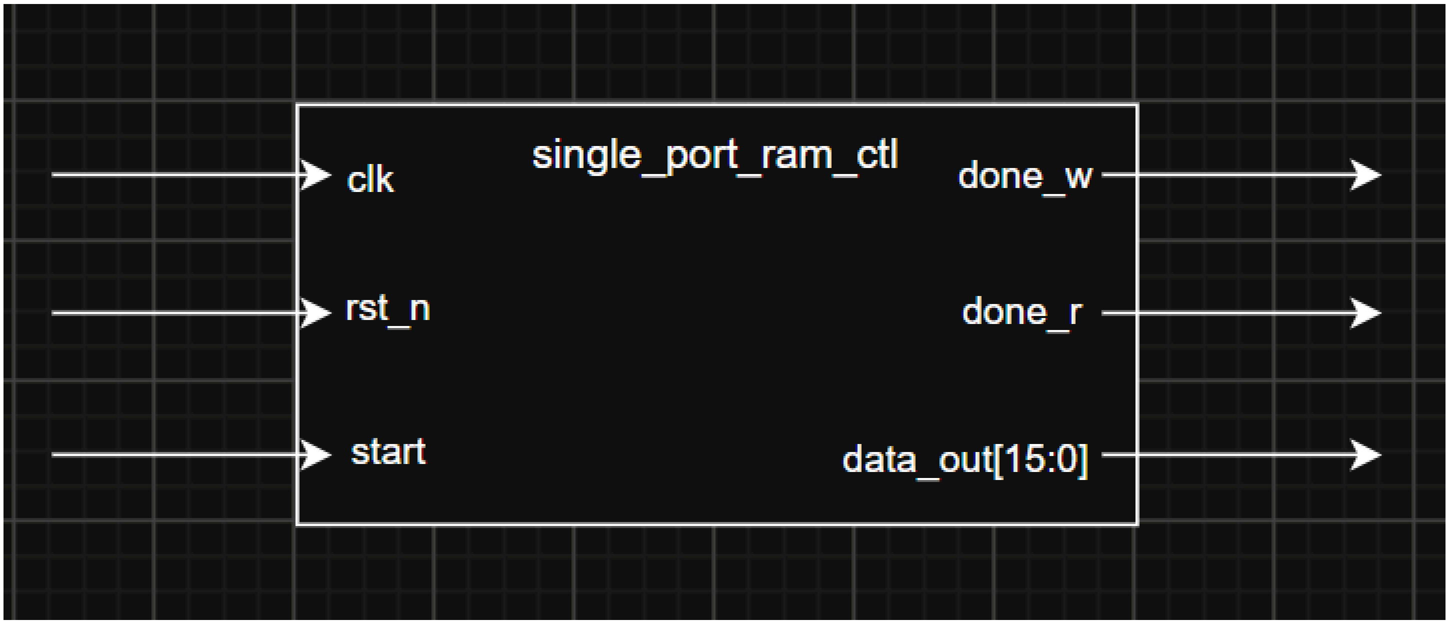
- read delay is 2 clk cycles
- addr is 0 to 99
- data input is interner register and the data is 1 to 100
- reset is async active low

Interface

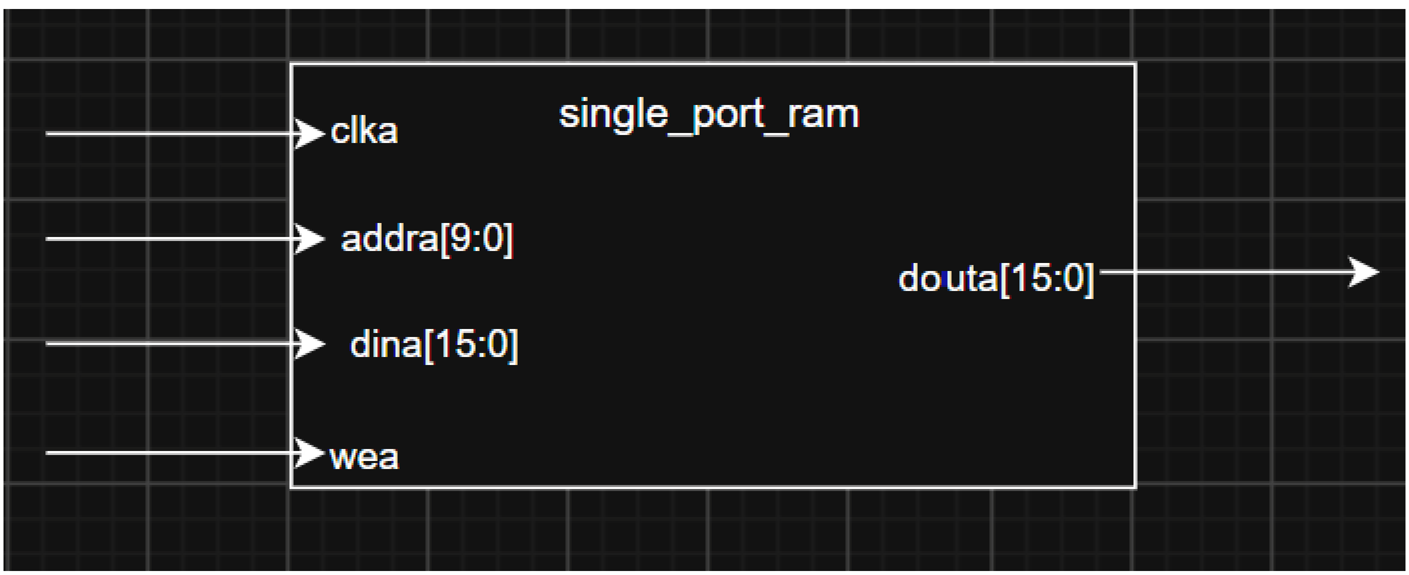
```
    input   logic               clk
,   input   logic               rst_n
,   input   logic               start

,   output  logic [15:0]        data_out
,   output  logic               done_w // 1clk pulse when write done
,   output  logic               done_r // 1clk pulse when read done
```

Block diagram

bd_top



bd_single_port_ram



Registers

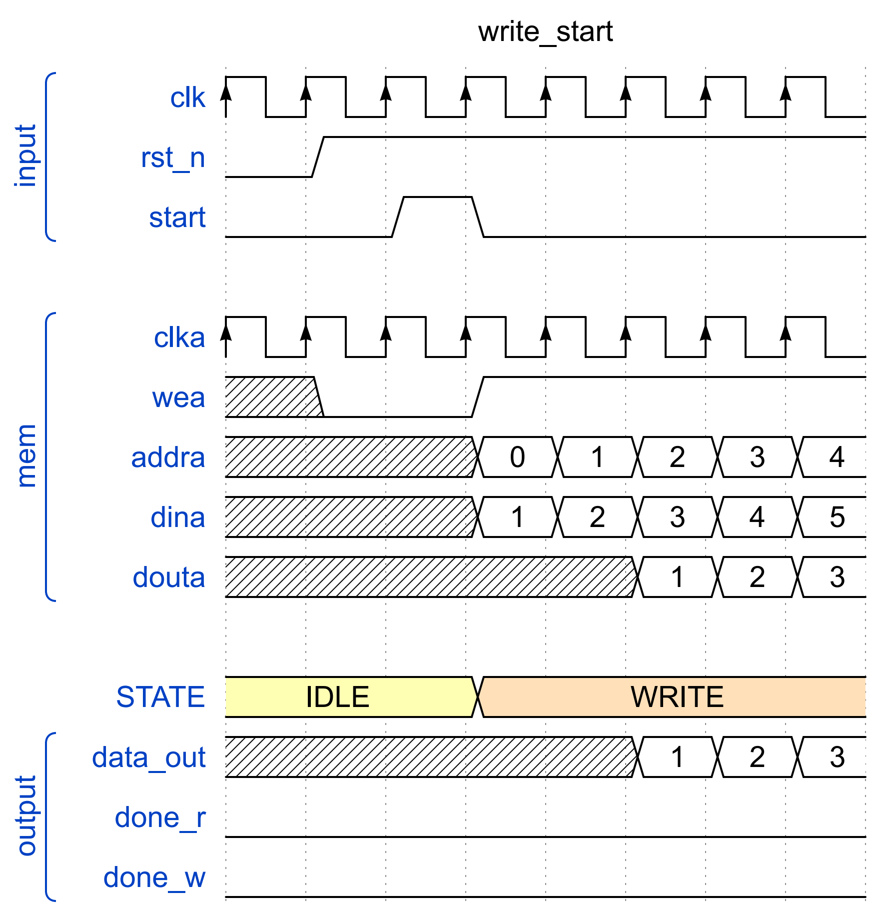
```
logic [15:0]            ram_data;
logic [9:0]             ram_addr;
```

RAM if

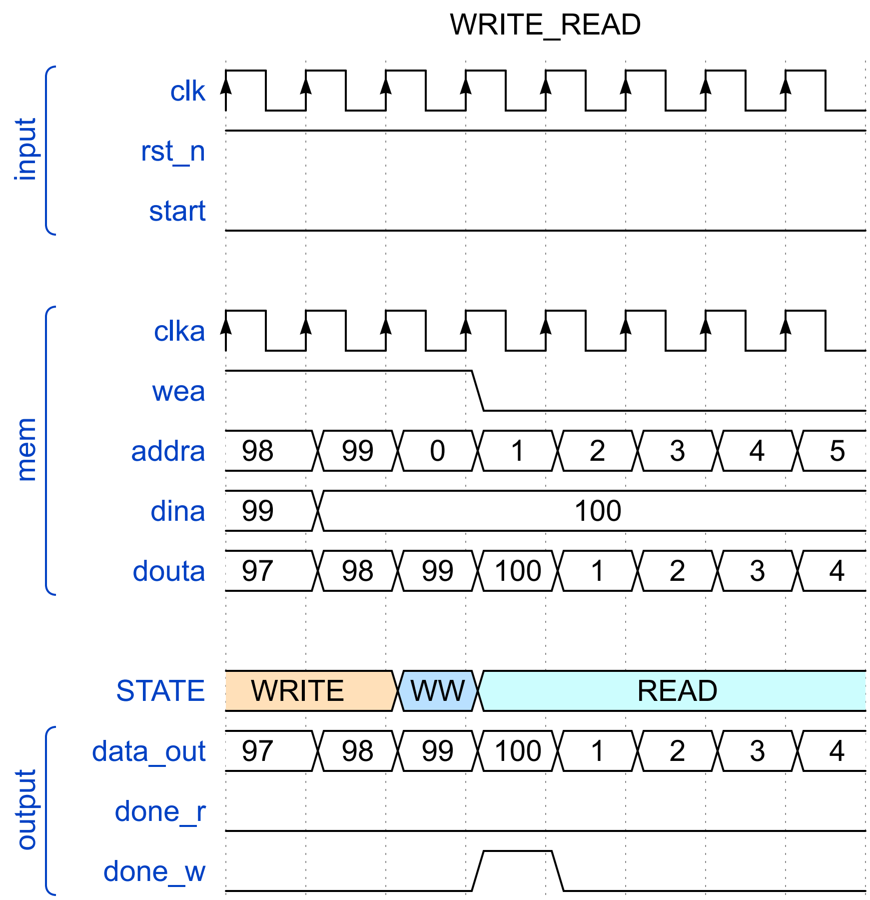
```
blk_mem_gen_0 u_blk_mem_gen_0 (
    .clka   (clk),
    .wea    (ram_we),
    .addra  (ram_addr),
    .dina   (ram_data),
    .douta  (data_out)
);
```

timing diagram

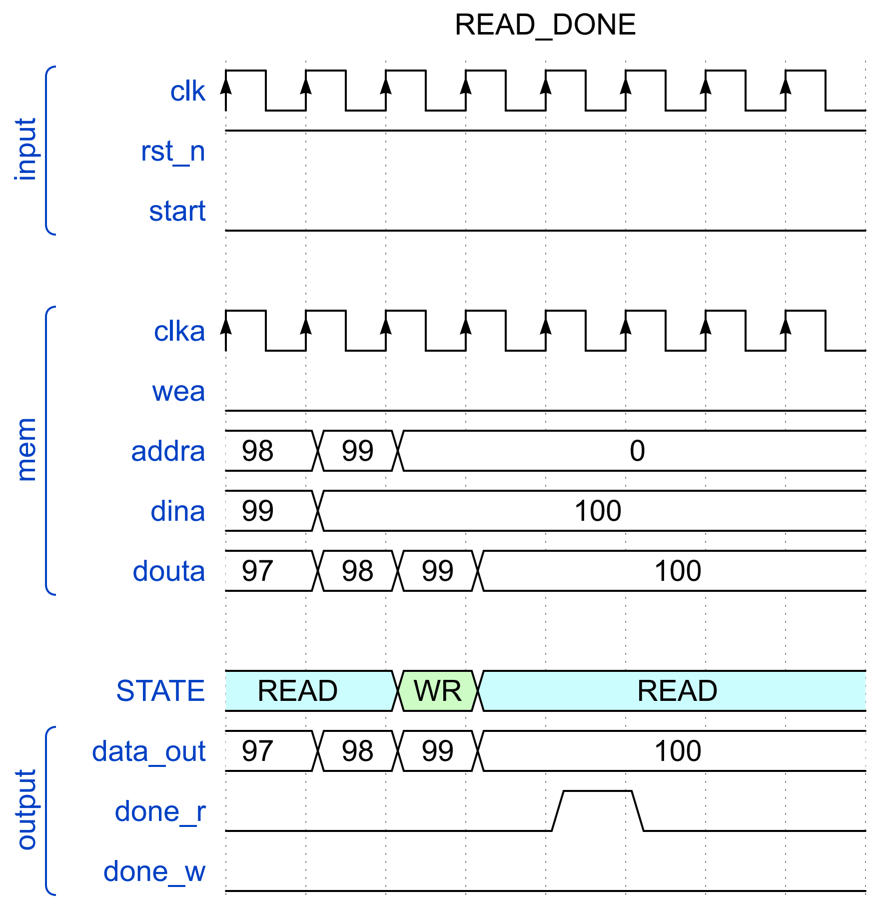
WRITE START



WRITE READ

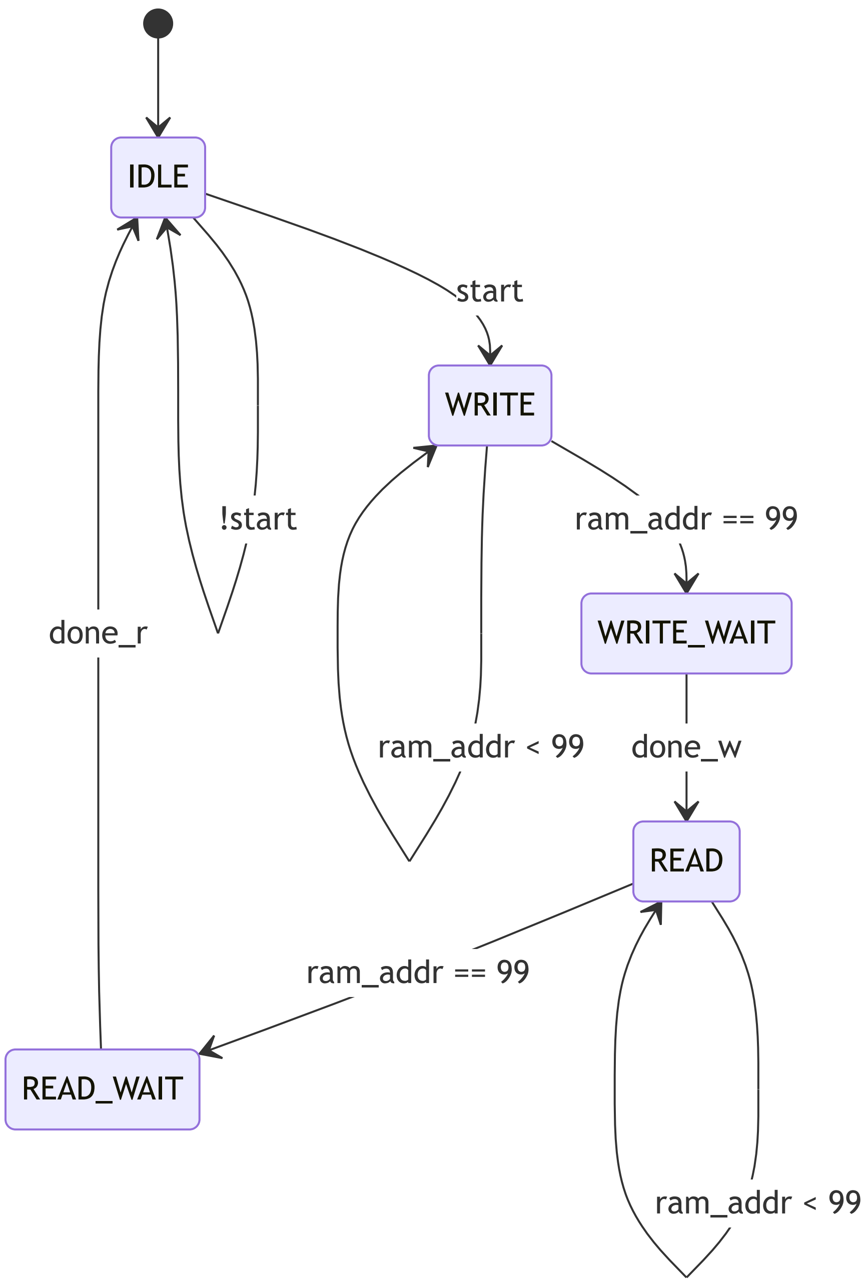

WRITE_READ
input
clk
rst_n
start
mem
clka
wea
addra
98
99
0
1
2
3
4
5
dina
99
100
douta
97
98
99
100
1
2
3
4
STATE
WRITE
WW
READ
output
data_out
97
98
99
100
1
2
3
4
done_r
done_w

READ DONE



READ_DONE
input
clk
rst_n
start
mem
clka
wea
addra
98
99
0
dina
99
100
douta
97
98
99
100
STATE
READ
WR
READ
output
data_out
97
98
99
100
done_r
done_w

STATE DIAGRAM



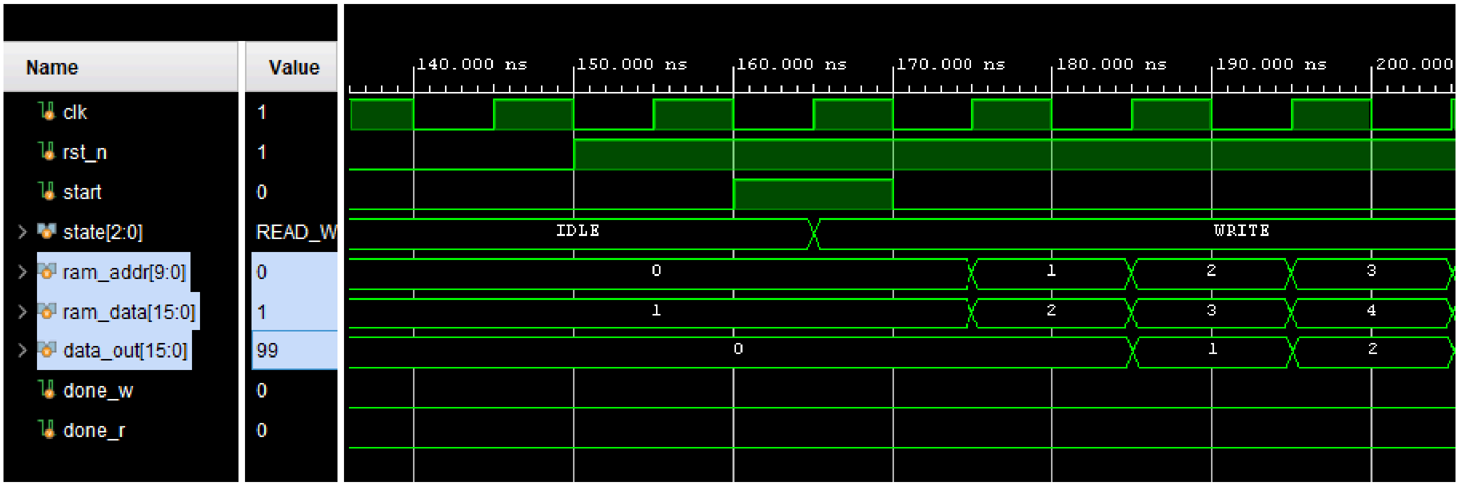
Test

Test scenario

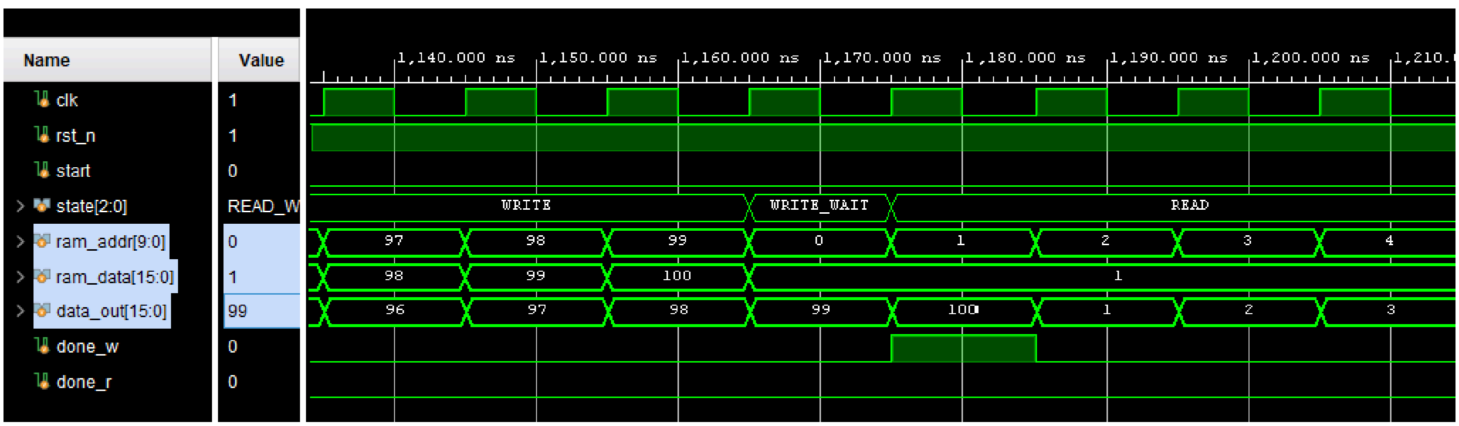
1. repeat(15) @posedge clk; // po sim때문에 10클럭 넘기고 수행
2. reset
3. start
4. wait(done)

be_sim

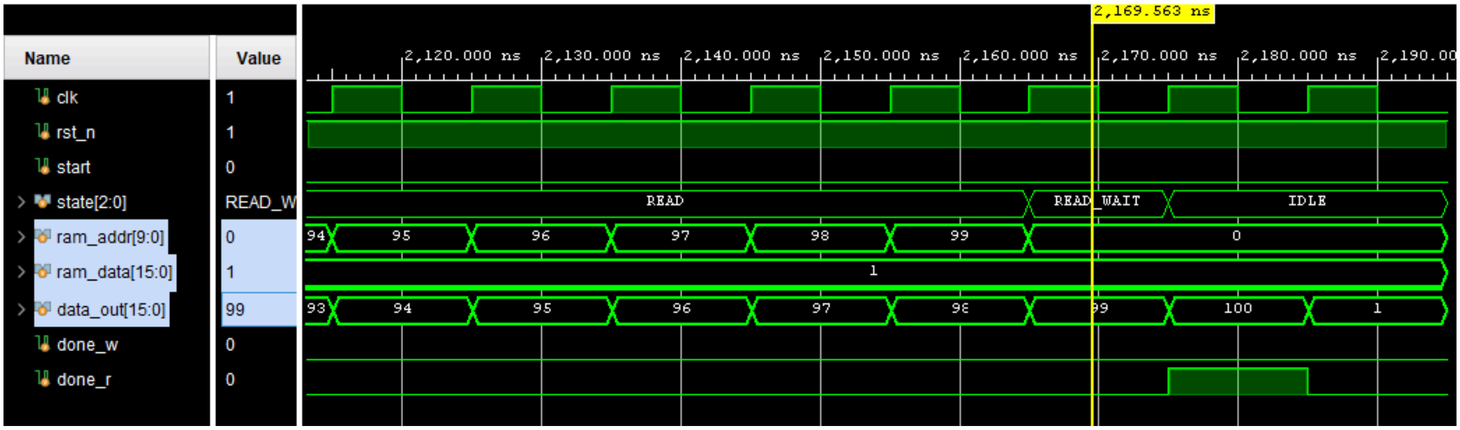
be_sim_write_start



be_sim_write_read

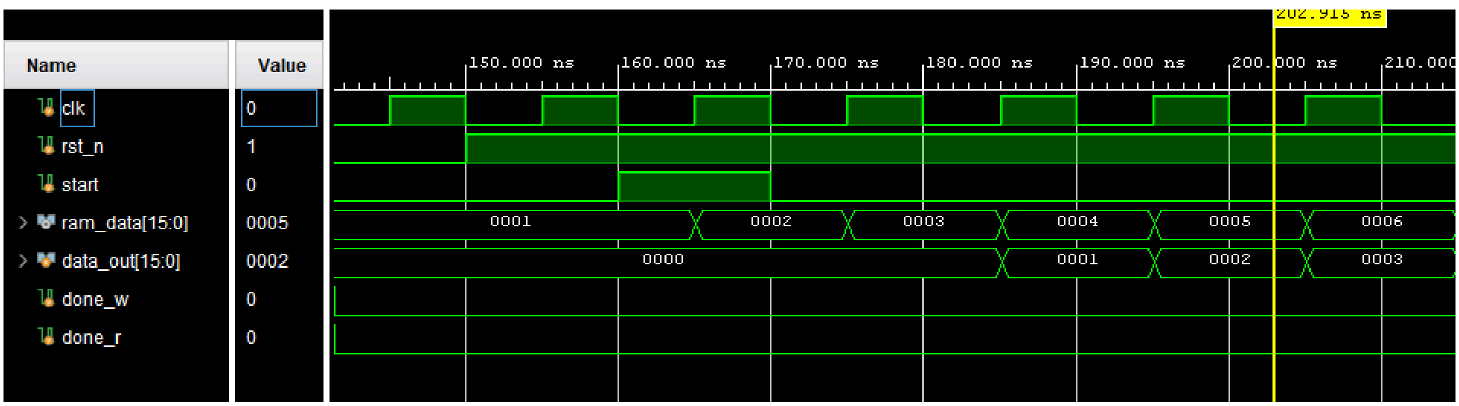


be_sim_read_end

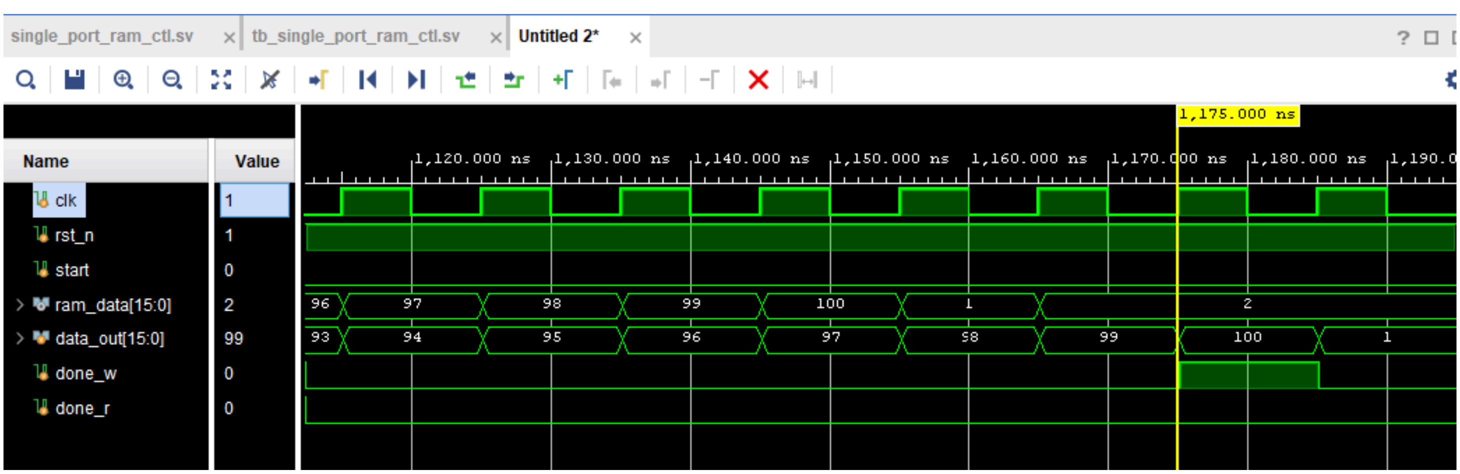


po_sim

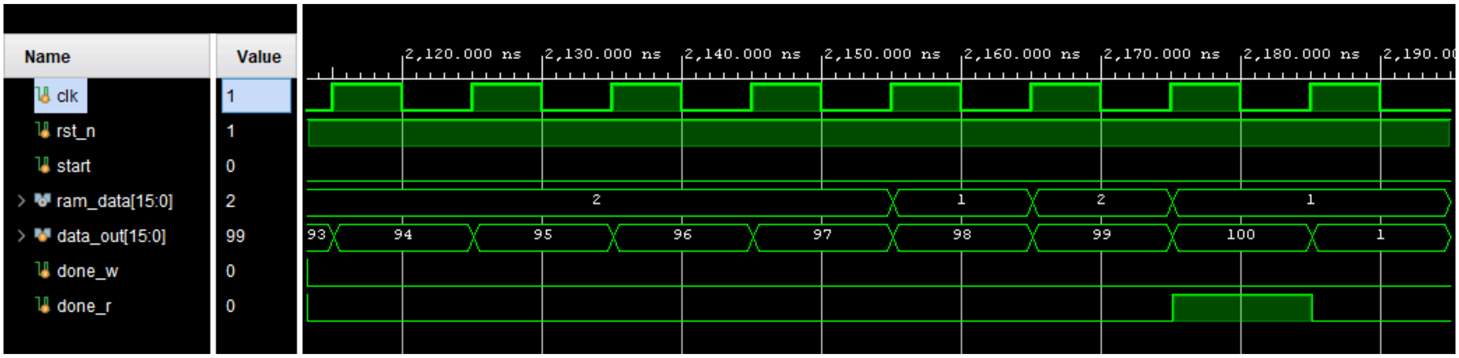
po_sim_write_start



po_sim_write_read



po_sim_read_end



Trouble shooting